

Ethanol Blending in Gasoline – Czechia

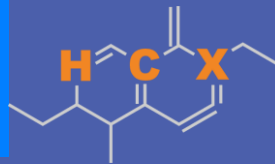
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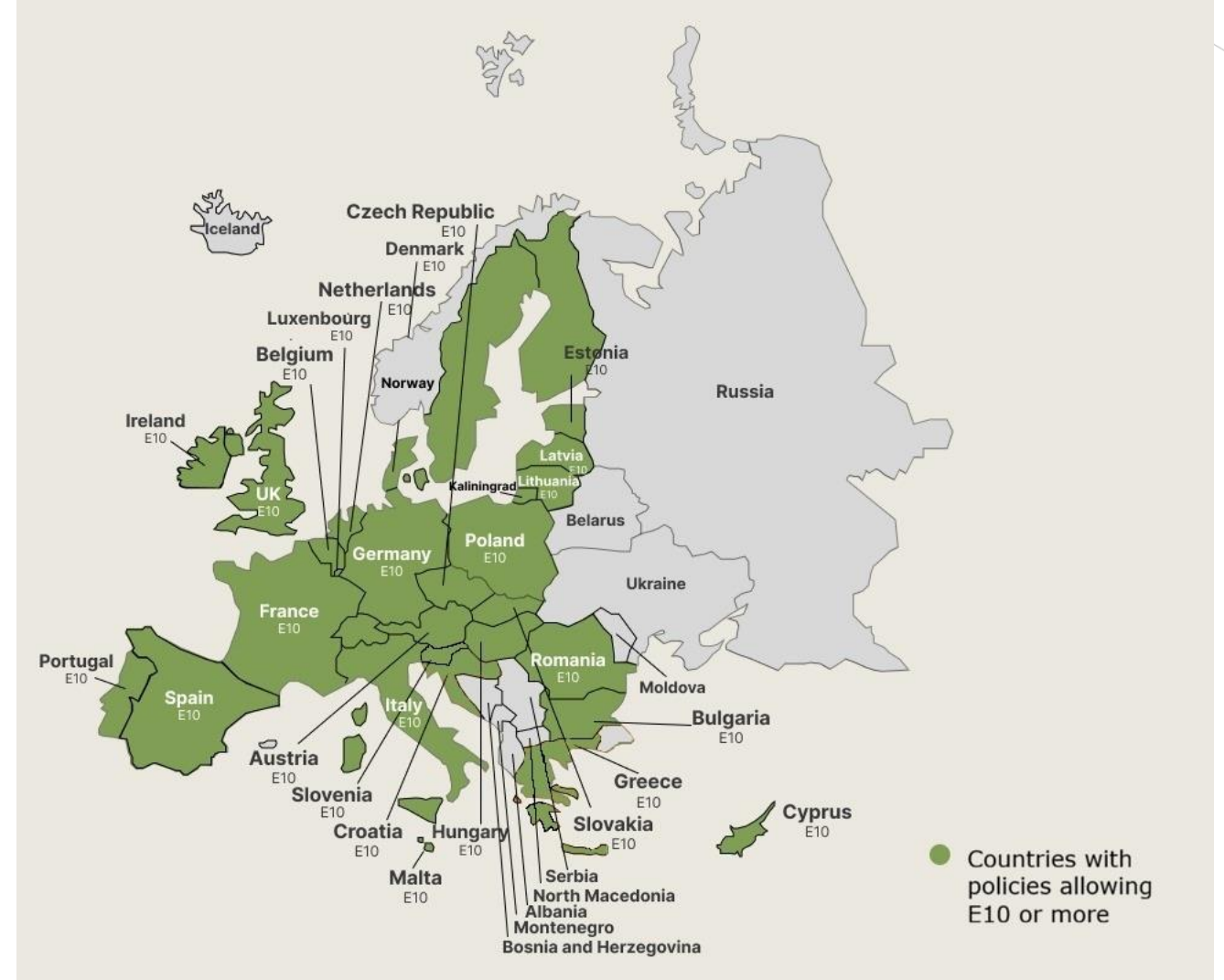


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Ethanol Blending in Europe

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

Ethanol Gasoline Blending in Czechia



In 2024, gasoline consumption in Czechia was 2,347 million liters (620 million gallons) and growth rate was 1.7%. Gasoline production and net imports were 2,328 and -936 million liters (615 and -247 million gallons) correspondingly. Czechia complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.1 RON.

In 2024, ethanol consumption in Czechia was 96 million liters (25 million gallons) representing 4.1% of gasoline demand. Ethanol production and Net Imports were 76 and 21 million liters (25 and 5 million gallons) correspondingly. Ethanol sources are Corn 50% and Sugar Beet 50%.

Source: Czechia Statistics, Eurostat

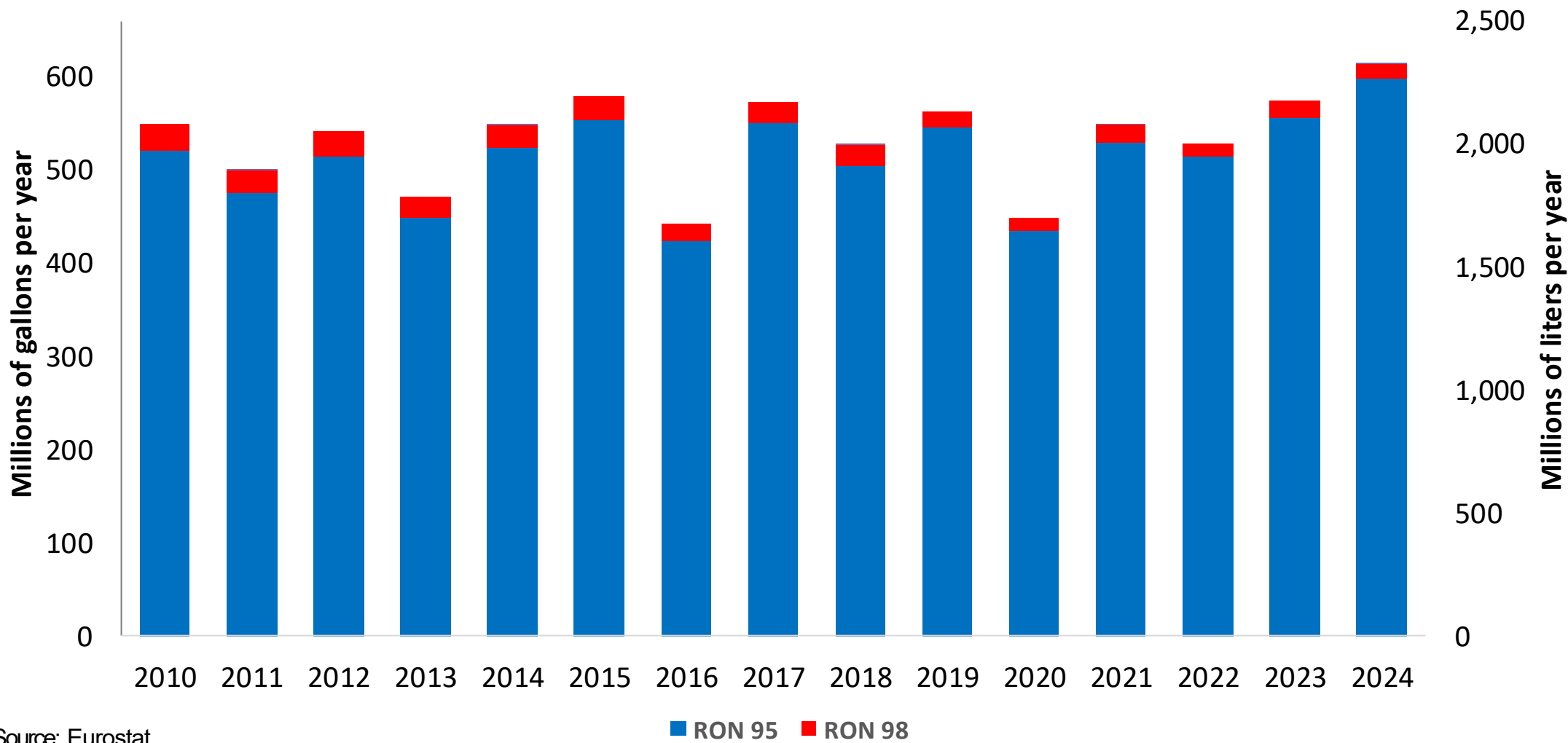
This stacked bar chart illustrates the annual consumption of RON 95 and RON 98 gasoline in the European Union from 2010 to 2024. The left vertical axis measures consumption in millions of gallons per year, ranging from 0 to 600. The right vertical axis measures consumption in millions of liters per year, ranging from 0 to 2,500. The horizontal axis lists the years from 2010 to 2024. Each bar is composed of two segments: a blue segment representing RON 95 and a red segment representing RON 98. The total height of each bar corresponds to the combined consumption of both grades. The data shows a general decline in total consumption from 2010 to 2020, followed by a steady increase through 2024. RON 95 consistently accounts for the vast majority of the total consumption, while RON 98 represents a smaller, relatively stable portion.

Year	RON 95 (Millions of gallons per year)	RON 98 (Millions of gallons per year)	Total (Millions of gallons per year)
2010	630	40	670
2011	610	35	645
2012	570	30	600
2013	540	25	565
2014	535	25	560
2015	540	25	565
2016	555	30	585
2017	550	25	575
2018	550	25	575
2019	555	25	580
2020	505	20	525
2021	525	20	545
2022	550	20	570
2023	580	20	600
2024	610	20	630

Source: Eurostat

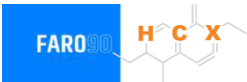
Gasoline Production in Czechia

Gasoline Production by Grade in Czechia

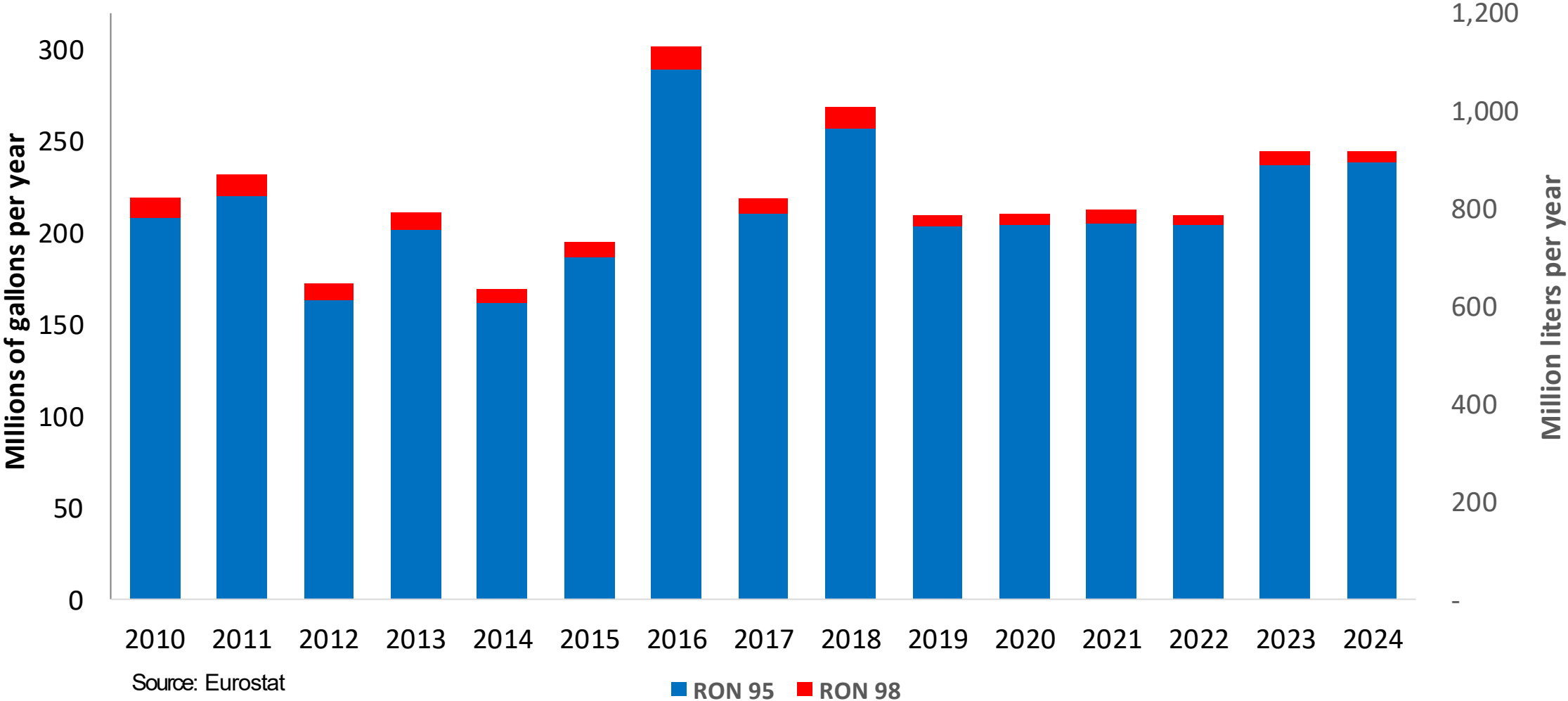


Source: Eurostat

Gasoline Imports in Czechia



Gasoline Imports by Grade in Czechia

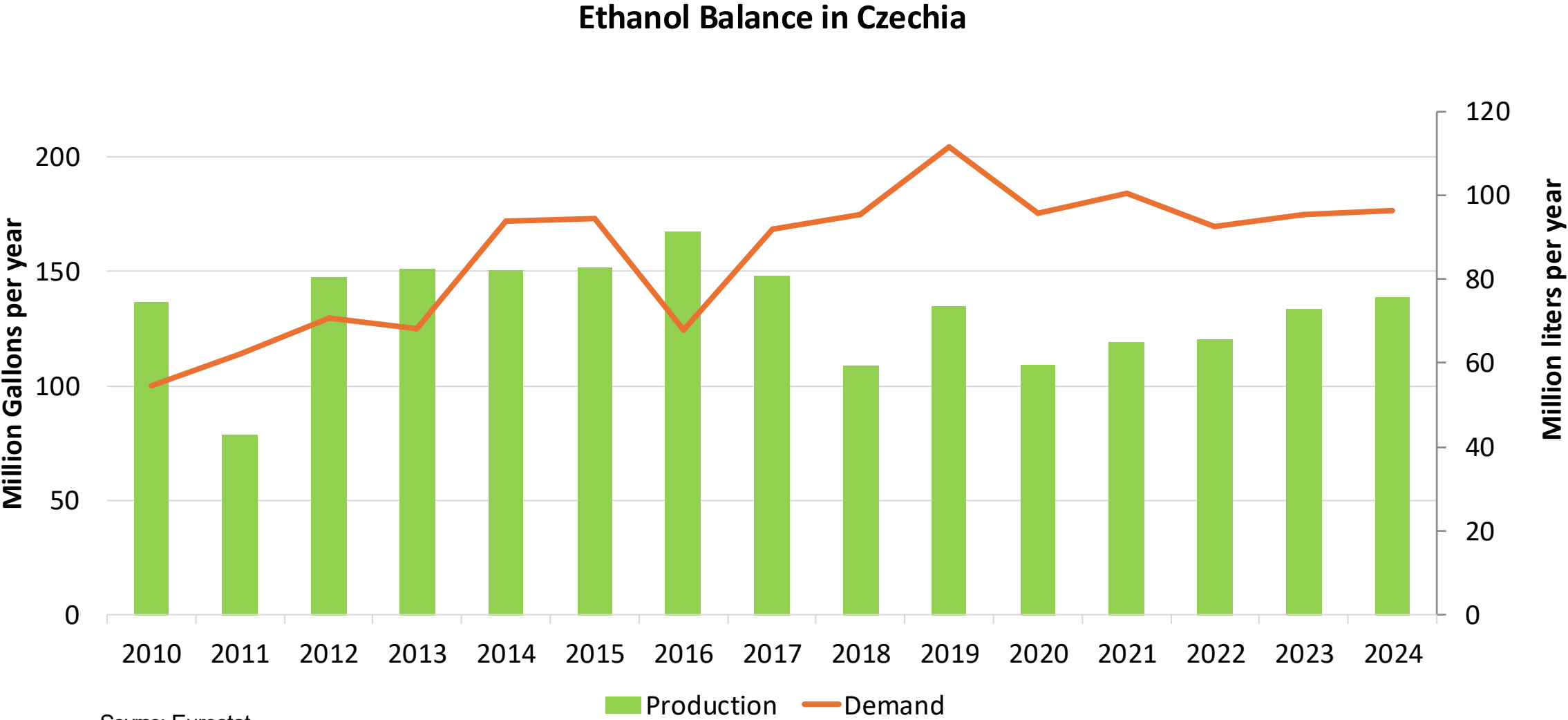


The FARO90 logo is shown in a blue box, followed by a chemical structure diagram of a substituted cyclohexene derivative. The structure features a six-membered ring with a double bond. Substituents include a methyl group, a propyl group, and a group labeled 'X'. The letters 'H', 'C', and 'X' are highlighted in orange in the original image.

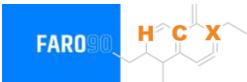
Source: Eurostat

Year	RON 95 (Millions of gallons per year)	RON 98 (Millions of gallons per year)	Total (Millions of gallons per year)	Total (Millions of liters per year)
2010	90	5	95	350
2011	108	6	114	435
2012	106	6	112	425
2013	118	7	125	475
2014	162	9	171	645
2015	210	10	220	840
2016	142	7	149	565
2017	220	10	230	875
2018	198	10	208	790
2019	212	7	219	840
2020	150	6	156	590
2021	218	9	227	870
2022	165	6	171	645
2023	215	7	222	845
2024	250	10	260	990

Ethanol Balance in Czechia



Gasoline Quality in Czechia



Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	1.1							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	6.0							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

The FARO90 logo is shown in a blue box, followed by a chemical structure diagram of a substituted cyclohexene derivative. The structure features a six-membered ring with a double bond. Substituents include a methyl group, a propyl group, and a side chain containing a carbonyl group and a methyl group. The letters H, C, and X are highlighted in orange on the structure.

Catalytic Gasoline
Light Primary Naphta
Heavy Primary Naphta
Reformate
Isopentane
Alkylate
Isomerate
Hydrocracked Gasoline
Coker Naphta
n-Butane
MTBE
ETBE
TAME
Aromatics

Country	Threat Category 1 (Dark Blue)	Threat Category 2 (Red)	Threat Category 3 (Grey)	Threat Category 4 (Purple)	Threat Category 5 (Light Grey)	Threat Category 6 (Orange)	Threat Category 7 (Yellow)	Threat Category 8 (Brown)	Threat Category 9 (Dark Blue)	Threat Category 10 (Light Yellow)	Threat Category 11 (Dark Purple)	Threat Category 12 (Dark Blue)
Austria	15%	15%	15%	15%	10%	10%	10%	10%	5%	5%	5%	5%
Belgium	18%	18%	18%	18%	10%	10%	10%	10%	5%	5%	5%	5%
Bulgaria	10%	10%	15%	20%	10%	10%	10%	10%	10%	5%	5%	5%
Croatia	15%	15%	15%	20%	10%	10%	10%	10%	10%	5%	5%	5%
Cyprus	15%	15%	15%	20%	10%	10%	10%	10%	5%	5%	5%	5%
Czechia	10%	10%	15%	20%	10%	10%	10%	10%	10%	5%	5%	5%
Denmark	15%	15%	15%	18%	10%	10%	10%	10%	5%	5%	5%	5%
Estonia	10%	10%	15%	20%	10%	10%	10%	10%	10%	5%	5%	5%
Finland	10%	10%	15%	20%	10%	10%	10%	10%	10%	5%	5%	5%
France	20%	20%	20%	18%	5%	5%	5%	10%	5%	5%	5%	5%
Germany	15%	15%	15%	18%	10%	10%	10%	10%	5%	5%	5%	5%
Greece	15%	15%	15%	20%	10%	10%	10%	10%	5%	5%	5%	5%
Hungary	12%	18%	15%	15%	5%	10%	10%	10%	10%	5%	5%	5%
Ireland	5%	15%	15%	25%	5%	15%	10%	5%	15%	5%	5%	5%
Italy	15%	15%	15%	20%	5%	5%	5%	10%	10%	5%	5%	5%
Latvia	10%	10%	15%	20%	10%	10%	10%	10%	10%	5%	5%	5%
Lithuania	10%	10%	15%	20%	10%	10%	10%	10%	10%	5%	5%	5%
Luxembourg	15%	15%	15%	18%	10%	10%	10%	10%	5%	5%	5%	5%
Malta	15%	15%	15%	20%	5%	5%	5%	10%	10%	5%	5%	5%
Netherlands	18%	18%	18%	18%	5%	10%	10%	10%	5%	5%	5%	5%
Poland	10%	10%	15%	20%	10%	10%	10%	10%	10%	5%	5%	5%
Portugal	15%	15%	15%	18%	5%	5%	5%	10%	5%	5%	5%	5%
Romania	10%	10%	15%	20%	10%	10%	10%	10%	10%	5%	5%	5%
Slovakia	10%	10%	15%	20%	10%	10%	10%	10%	10%	5%	5%	5%
Slovenia	15%	15%	15%	18%	10%	10%	10%	10%	5%	5%	5%	5%
Spain	15%	15%	15%	20%	5%	5%	5%	10%	5%	5%	5%	5%
Sweden	15%	15%	15%	18%	10%	10%	10%	10%	5%	5%	5%	5%
UK	5%	15%	15%	20%	5%	15%	10%	5%	15%	5%	5%	5%

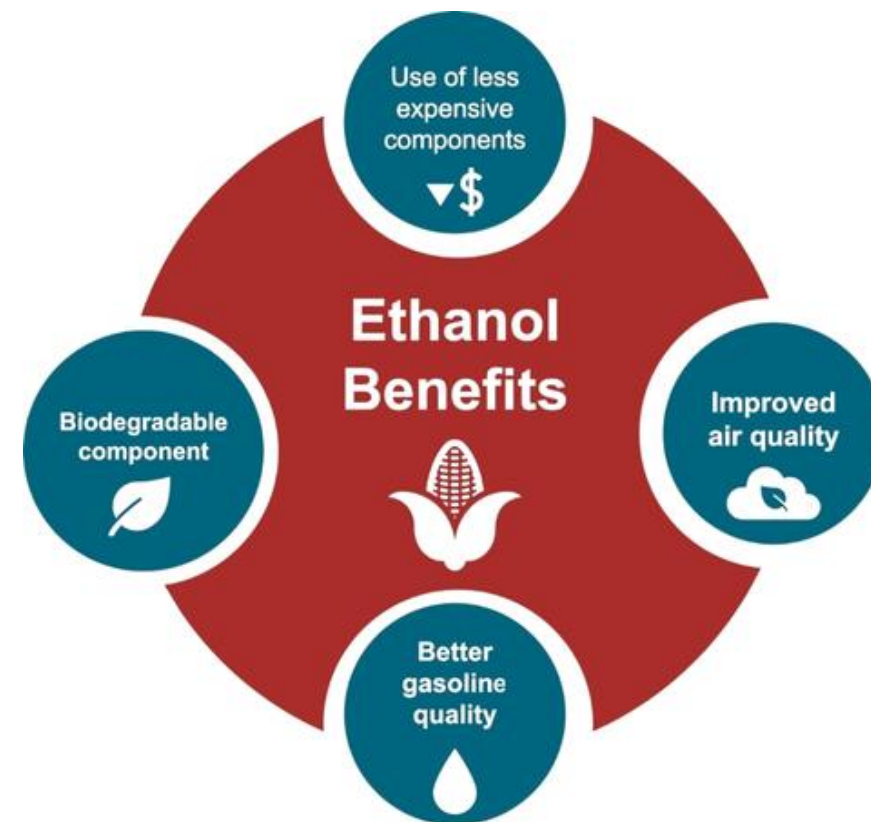
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

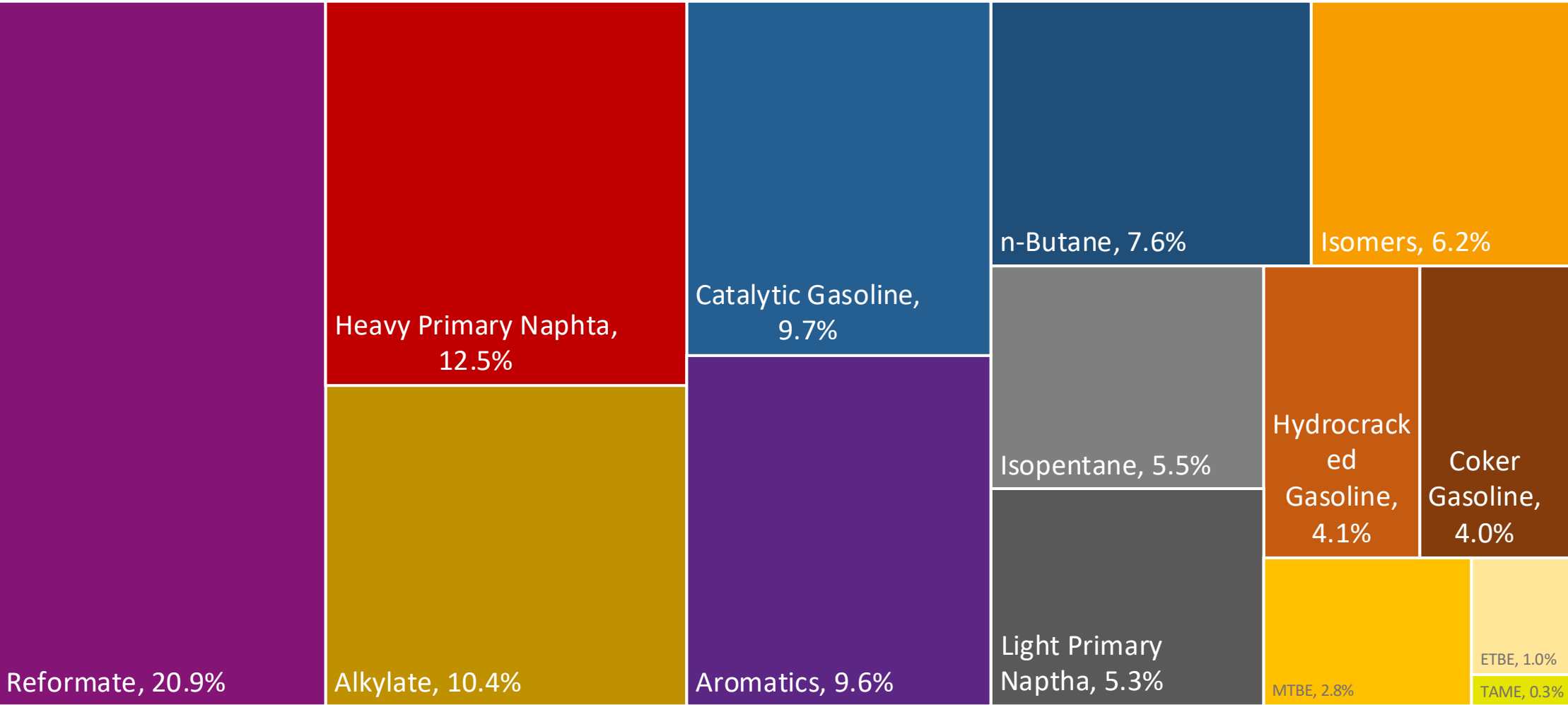
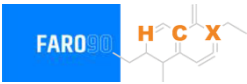
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

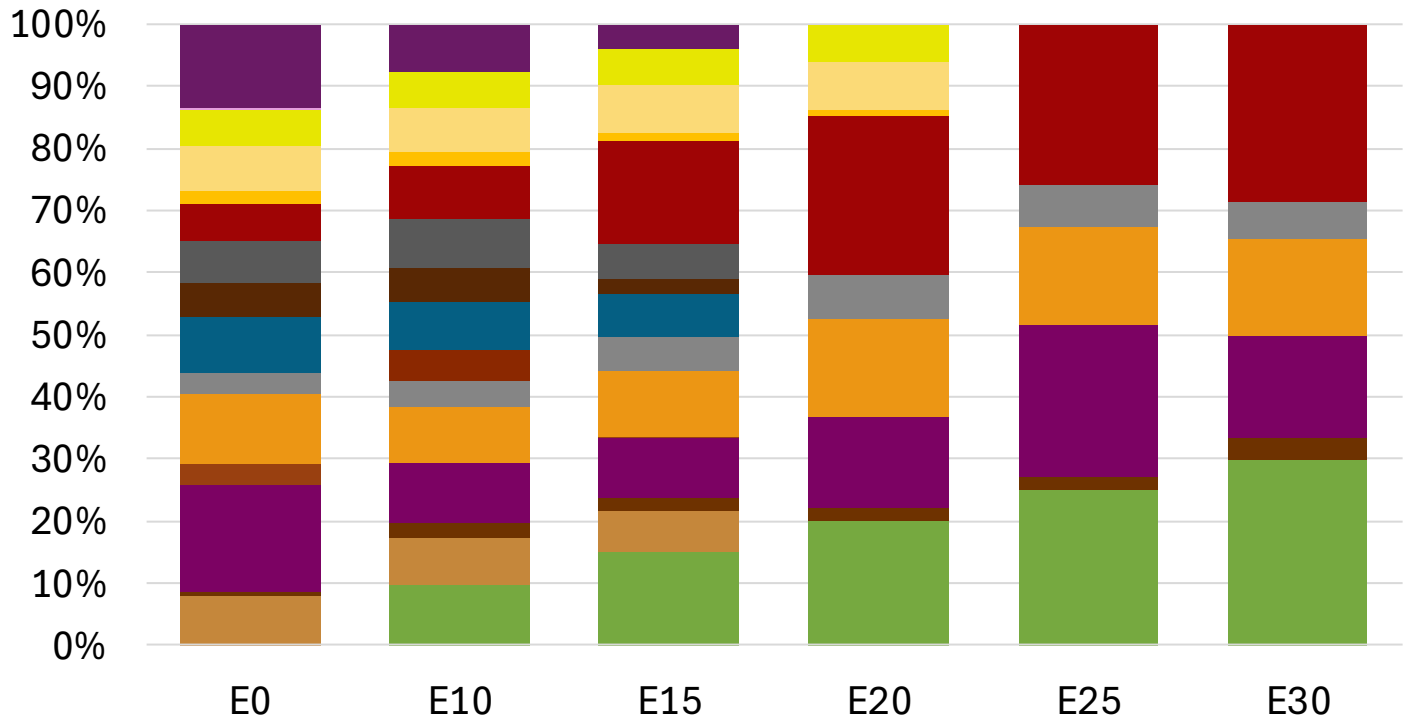
The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Czechia Available Blending Components



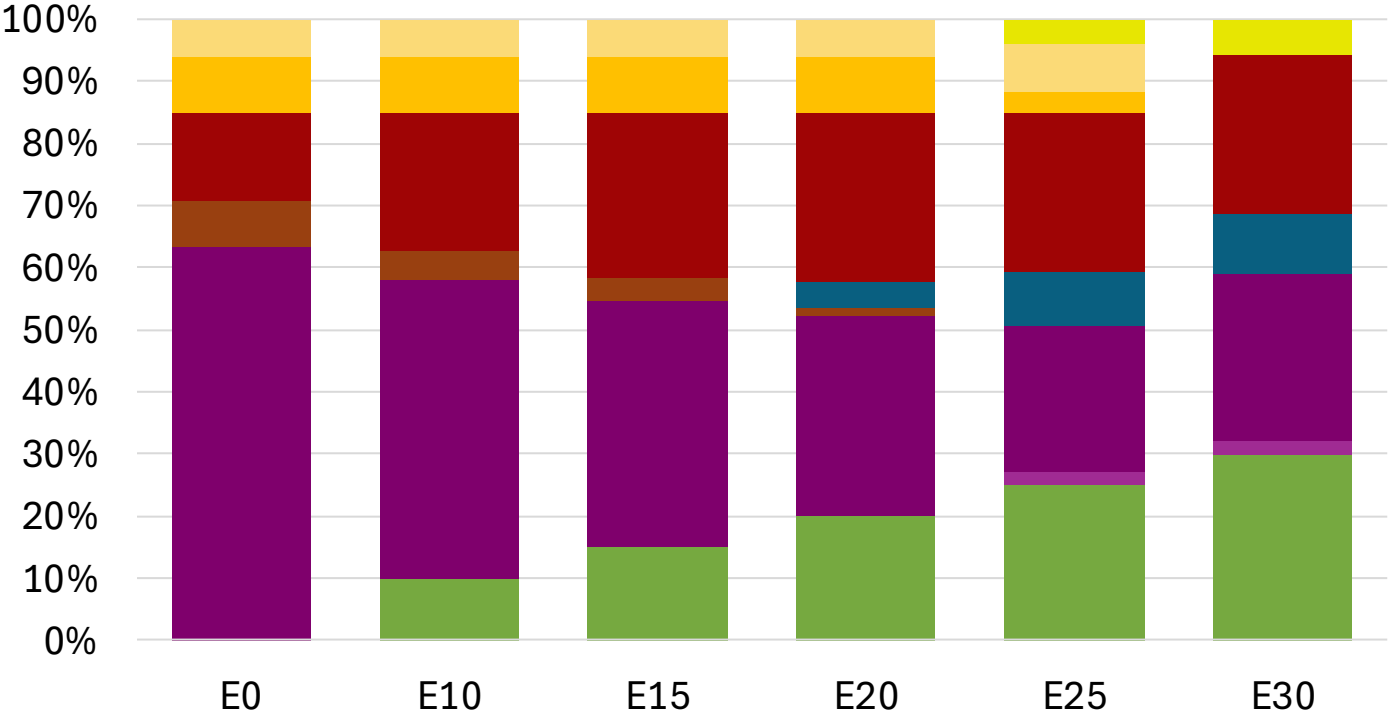
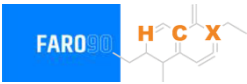
Czechia – Gasoline 95 – Constant Octane



Average CIF 2024
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US\$/gal)	\$ 2.314	\$ 2.176	\$ 2.076	\$ 1.961	\$ 1.915	\$ 1.862

Czechia - Gasoline 98 - Constant Octane

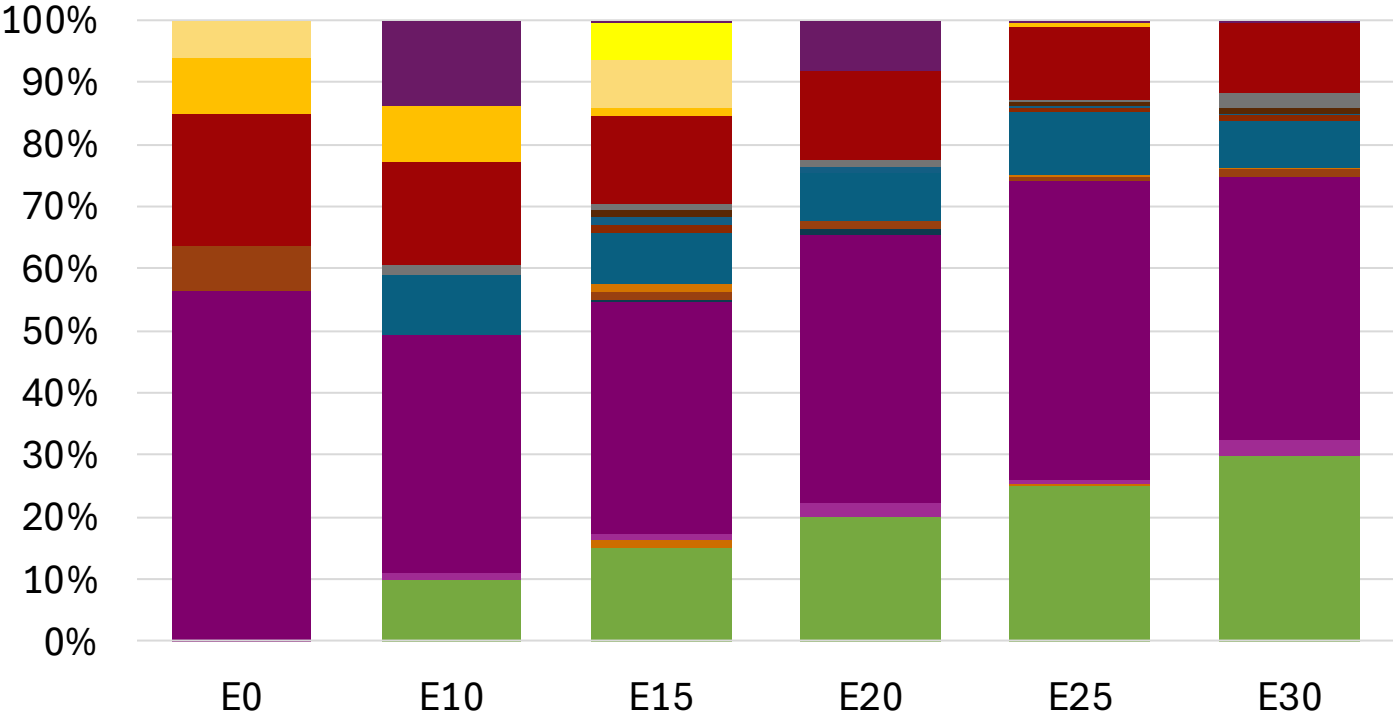


Average CIF 2024
Prices

Octane (RON)	98.0					
Price (US\$/gal)	\$ 2.334	\$ 2.217	\$ 2.148	\$ 2.076	\$ 2.011	\$ 1.961

Source: Faro90

Czechia – Gasoline 95 – Increased Octane

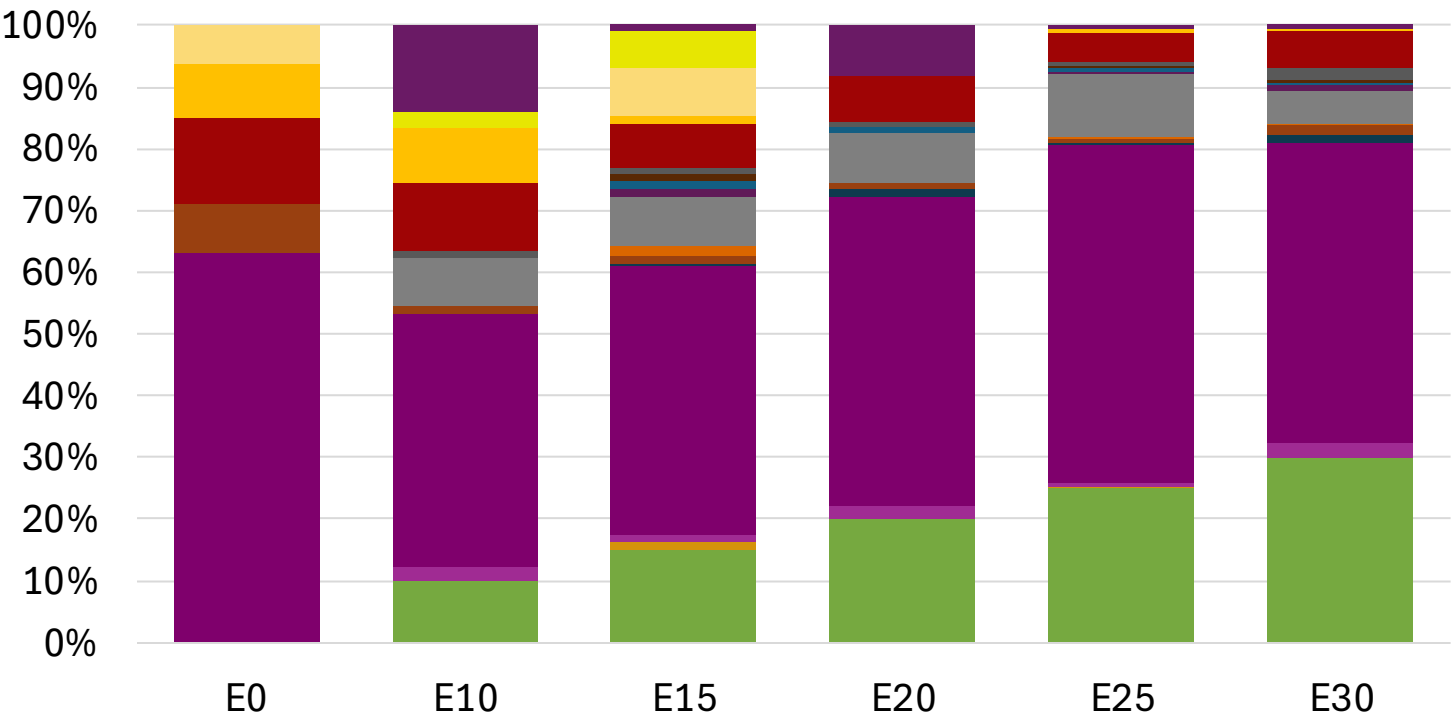


Average CIF 2024
Prices

Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US\$/gal)	\$ 2.202	\$ 2.202	\$ 2.202	\$ 2.202	\$ 2.202	\$ 2.202

Source: Faro90

Czechia – Gasoline 98 – Increased Octane



Average CIF 2024
Prices

Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US\$/gal)	\$ 2.334	\$ 2.334	\$ 2.334	\$ 2.334	\$ 2.334	\$ 2.334

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

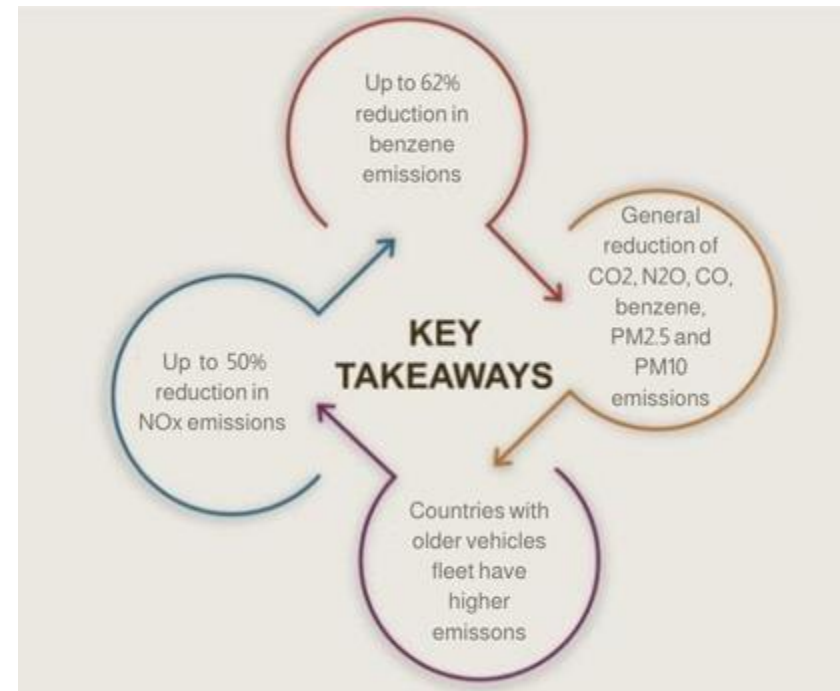
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

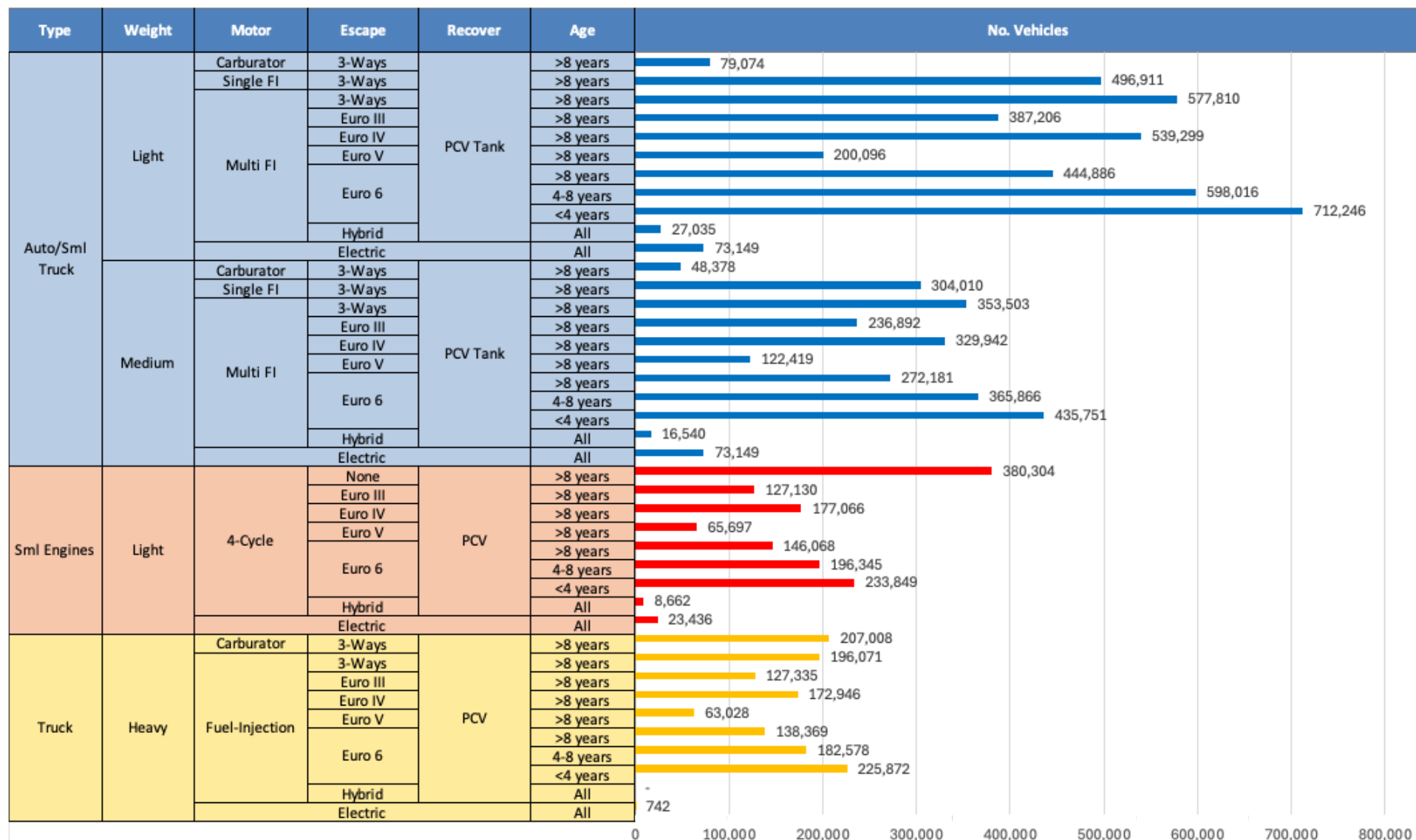
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

*Source: Bureau of transportation statistics.

Main Results



Gasoline Vehicle Fleet - Czechia

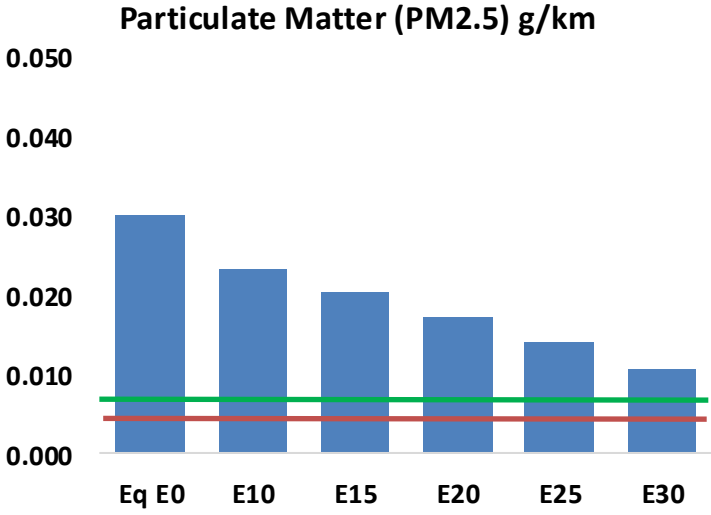
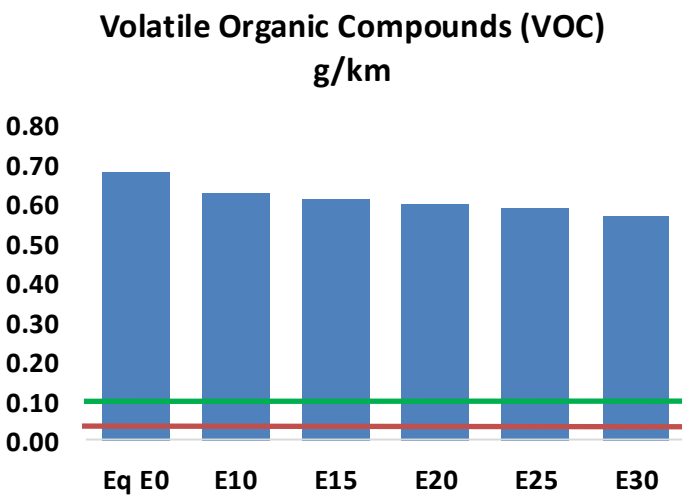
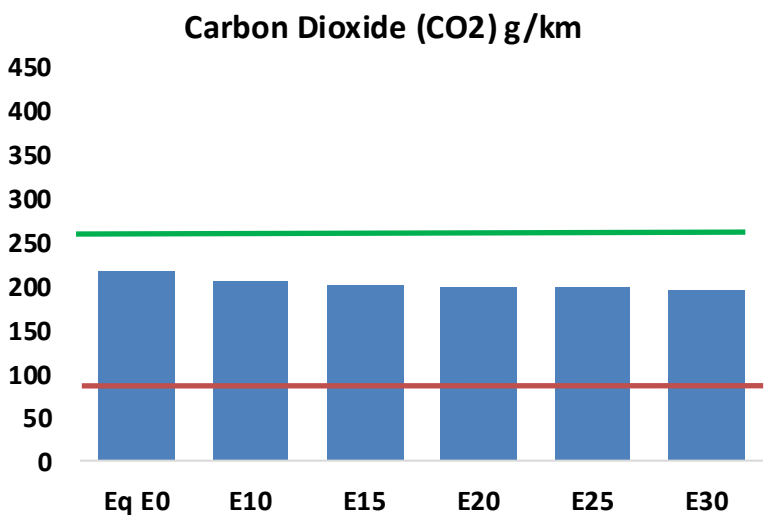
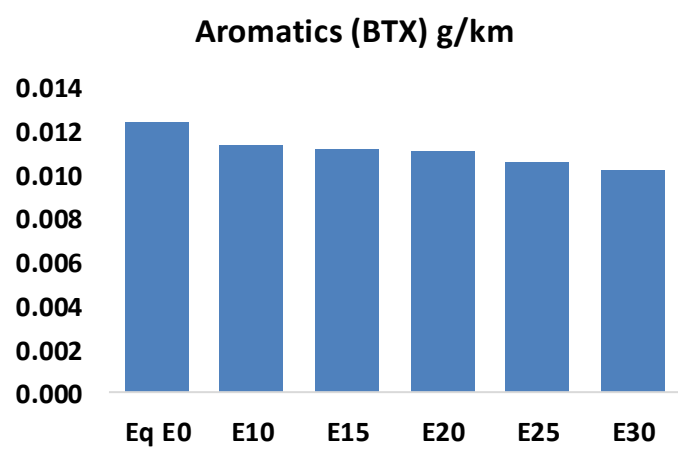
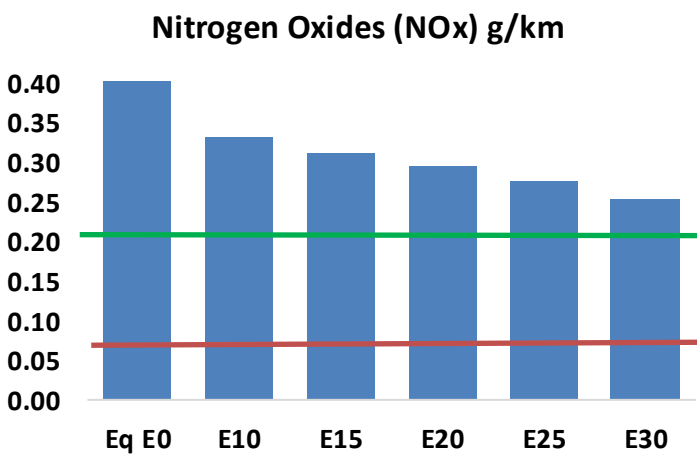
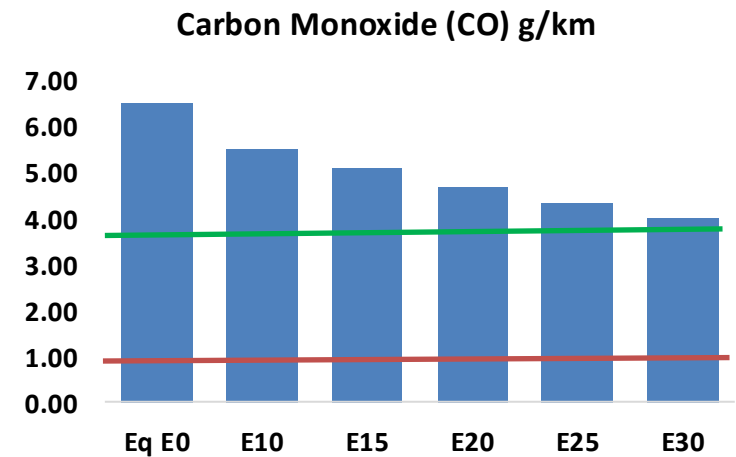
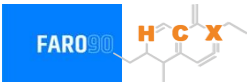


Vehicle Fleet: 9,366,865
Gasoline Vehicle Fleet: 3,468,160

Source: Eurostat, ACEA

Czechia – Vehicular Emissions

TIER III BIN 125 United States
Euro 6 Standard



Czechia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	514,816	475,239	435,661	403,229	370,591	345,495	316,625	-15%	-28%
CO2	17,030,957	16,593,018	16,179,409	15,854,118	15,693,151	15,618,037	15,330,151	-5%	-8%
VOC	54,073	51,964	49,856	48,528	47,375	46,544	45,240	-8%	-12%
NOx	37,565	28,173	26,295	24,783	23,373	21,817	20,172	-30%	-38%
SOx	196	181	166	154	142	129	115	-15%	-28%
NH3	5,868	5,810	5,752	5,779	5,845	5,890	5,928	-2%	0%
N2O	308	306	305	306	313	316	320	-1%	2%
CH4	11,875	11,875	11,875	12,053	12,314	12,540	12,754	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	361	343	324	310	297	288	274	-10%	-18%
Acetaldehyde	771	885	1,299	1,977	2,694	3,079	3,638	68%	249%
Formaldehyde	2,546	2,475	2,885	3,388	3,549	3,926	4,274	13%	39%
Benzene	987	948	898	884	877	839	812	-9%	-11%
PM2.5	1,023	909	794	689	585	475	360	-22%	-43%
PM10	1,353	1,202	1,050	912	774	628	476	-22%	-43%

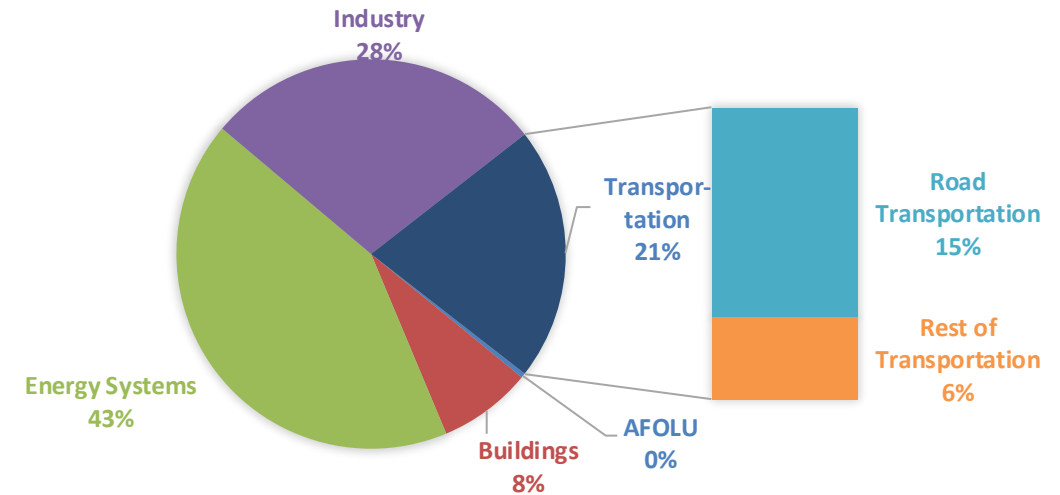
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

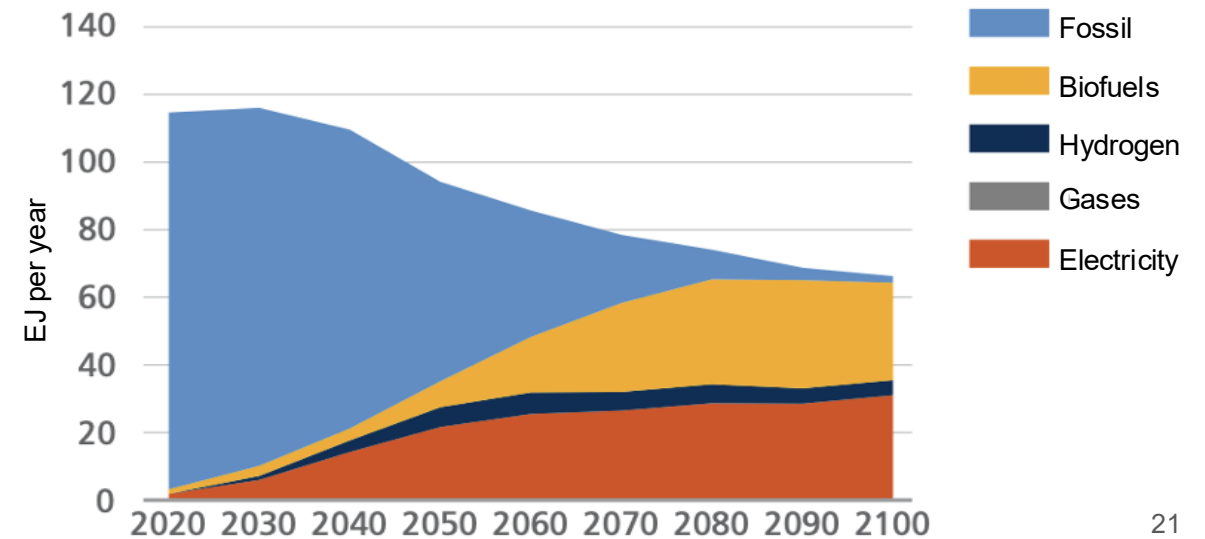
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

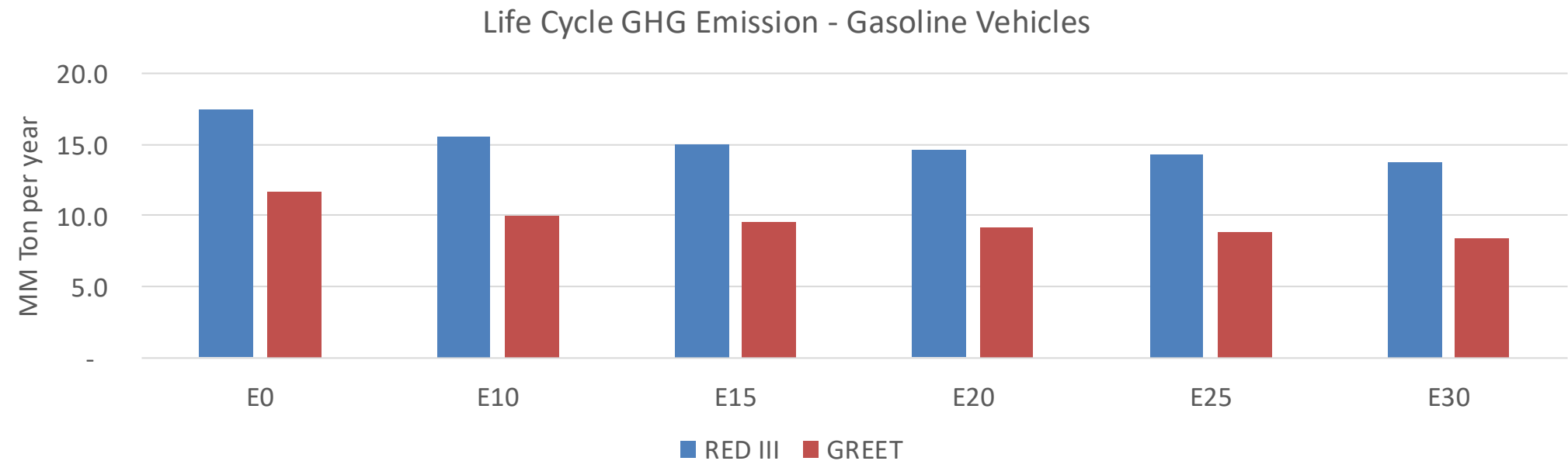
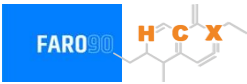
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Czechia – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	108.0
Target @ 2035 (MMT/year)	60.8
Delta	47.2

%Reduction vs E0	E0	E10	E15	E20	E25	E30
REET 2024	0.0%	15.1%	18.6%	21.5%	24.3%	27.9%
RED II/III	0.0%	11.1%	13.8%	16.2%	18.4%	21.2%

% Target@2035	E0	E10	E15	E20	E25	E30
REET 2024	0.0%	3.8%	4.6%	5.3%	6.0%	6.9%
RED II/III	0.0%	4.1%	5.1%	6.0%	6.8%	7.8%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90